

Zeros That Hold a Place

Answer Key

Grades 2–3

Quick Answers

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|--------|--------|------------------|
| 1. 407 | 4. 308 | 7. 308, 380, 830 |
| 2. 605 | 5. 720 | 8. 700, 70, — |
| 3. — | 6. 349 | |
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Curriculum

Level: Grades 2–3

2.NBT.A – problems 1, 2, 3, 4, 5, 6, 7, 8

Difficulty 1–4 of 5 across 12 tagged checks.

Common wrong answers

4. *If they got 380: slid the 8 into the tens place because the tens digit was 0, writing $300 + 80$*
5. *If they got 702: counted the 2 as 2 ones instead of 2 tens because the ones digit was 0, writing $700 + 2$*
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1. Chart: 4 hundreds, 0 tens, 7 ones. Read the chart one place at a time: 4 hundreds is 400, 0 tens is 0, and 7 ones is 7. Expanded form: $400 + 0 + 7$. Adding the pieces gives 407. The 0 is what keeps the 7 in the ones place — without it the 7 would slide over and the number would read 47.

2. Expanded form to standard form: $600 + 0 + 5$. The pieces name the places: 6 hundreds, 0 tens, 5 ones. Write one digit per place: 6 in the hundreds place, 0 in the tens place, 5 in the ones place, which gives 605. A common wrong answer here is 65 — that is what happens when the 0 is left out and the 5 slides into the tens place.

3. Compare 406 and 460. Both numbers have 4 hundreds, so the hundreds place does not decide it. Look at the tens: 406 has 0 tens, and 460 has 6 tens. 0 tens is less than 6 tens, so $406 < 460$. Same digits, different places — the 0 is doing real work.

4. Find the mistake (Jamal). (a) Jamal skipped the tens place. Because the tens digit is 0, he moved the 8 over into the tens place and wrote 80 instead of 8, so his pieces make 380. [Manual review: any wording that says the 8 belongs in the ones place is correct.]

(b) Correct expanded form: $300 + 0 + 8$. Adding the pieces gives 308. Check the places: 3 hundreds, 0 tens, 8 ones.

5. Find the mistake (Ruby). (a) Ruby read the 2 as 2 ones. In 720 the 2 sits in the *tens* place, so it is worth 20, not 2. The 0 in the ones place is what pushes the 2 into the tens place, and her answer of 702 puts the 2 back in the ones place. [Manual review: the explanation must name the tens place.]

(b) Correct expanded form: $700 + 20 + 0$. Adding the pieces gives $\boxed{720}$.

6. Nina's blocks. Start: 3 hundreds, 0 tens, 9 ones, which is $300 + 0 + 9 = 309$. The teacher adds 4 tens, which is 40, so the tens place changes from 0 to 4. Now she has 3 hundreds, 4 tens, 9 ones: $300 + 40 + 9$. Nina's blocks show $\boxed{349}$. Notice that only the tens digit changed — the hundreds and ones stayed put because each place holds its own value.

7. Order 380, 830, 308 from least to greatest. Compare the hundreds first: 380 and 308 both have 3 hundreds, and 830 has 8 hundreds, so 830 is the greatest. Now compare 380 and 308 in the tens place: 8 tens versus 0 tens, so 308 is the smallest. Least to greatest: $\boxed{308, 380, 830}$. All three use the same three digits; only the places are different.

8. Challenge: 7 hundreds versus 7 tens. (a) 7 hundreds, 0 tens, 0 ones is $700 + 0 + 0 = \boxed{700}$. 0 hundreds, 7 tens, 0 ones is $0 + 70 + 0 = \boxed{70}$.

(b) The same digit 7 is worth 700 in one number and 70 in the other, because of the place it sits in, so $\boxed{700 > 70}$. The zeros are what hold the 7 in its place: erasing them would leave the same digit 7 in both numbers.